SynthMicro — Whole Blood Smear XAI Report

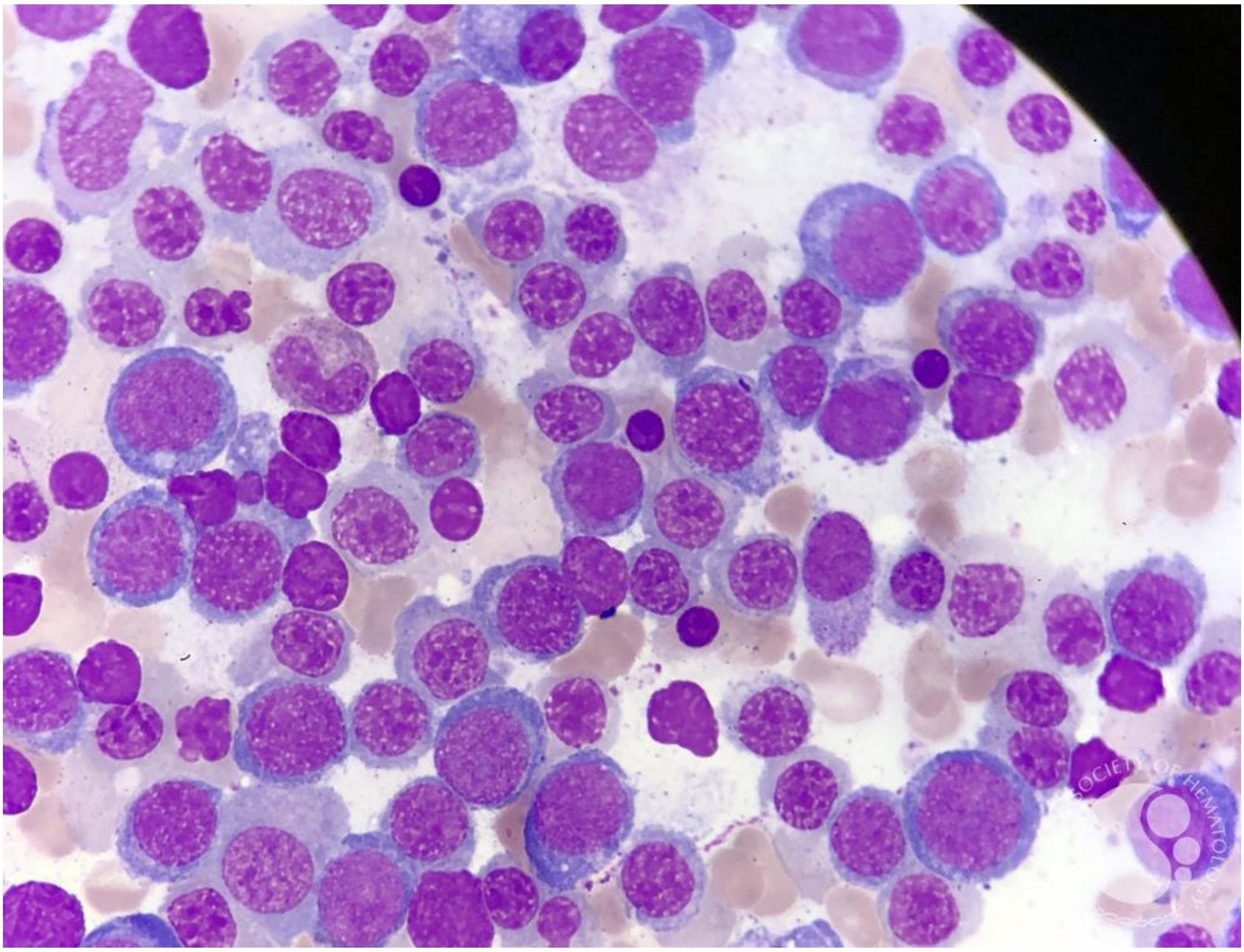
Analysis ID: 4ac063a3e4d946708fc5ae99efab886b
Generated: 2026-09-08 01:03:15

Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	115
Nucleated-cell candidates	115
Myeloblast-like candidates	66
Myeloblast-like proportion	57.39%

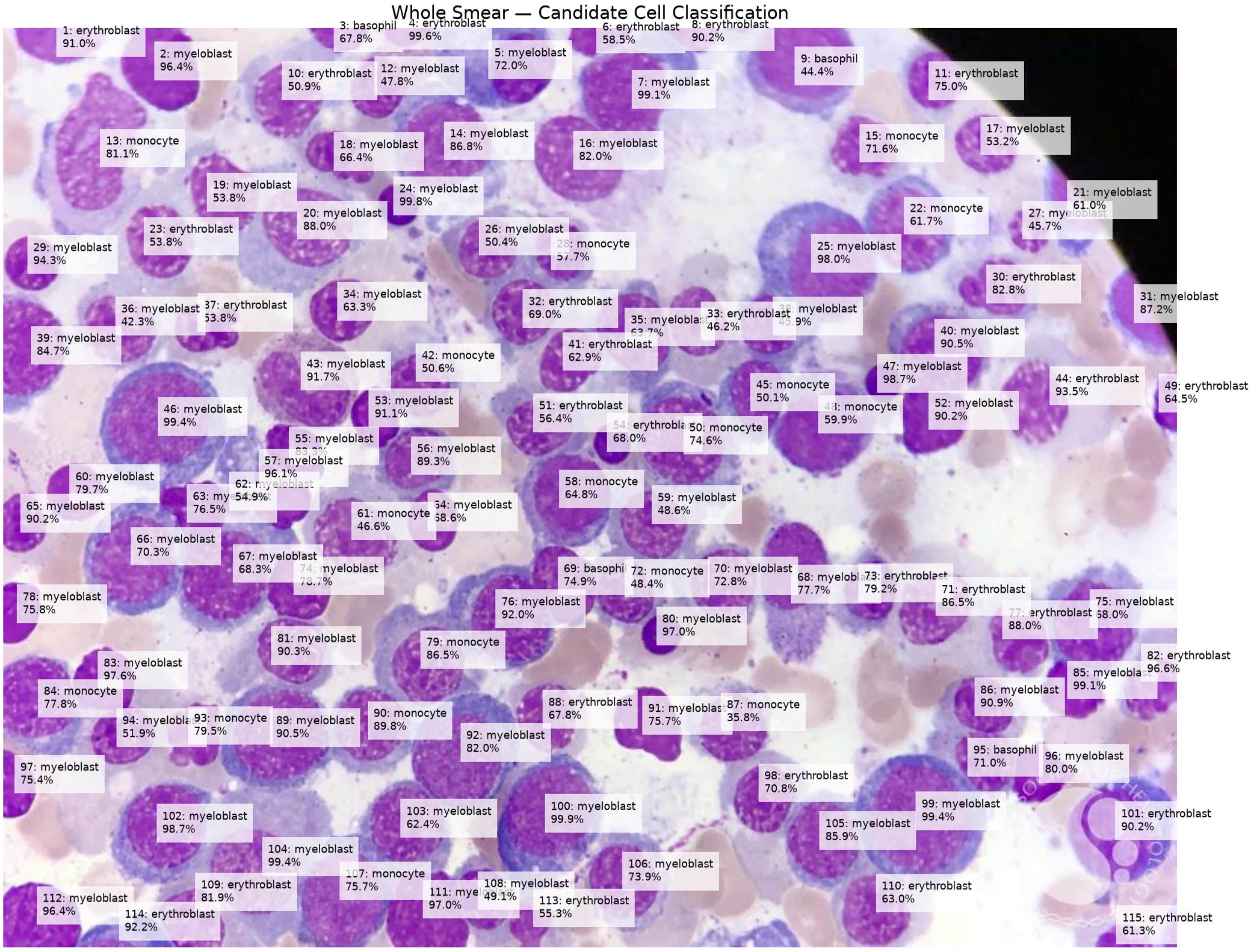
Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay



Cell-Level Results

ID	Area	Purple	Class	Confidence
1	1819	86.1	erythroblast	91.0%
2	3793	109.3	myeloblast	96.4%
3	783	91.8	basophil	67.8%
4	593	96.4	erythroblast	99.6%
5	5923	70.7	myeloblast	72.0%
6	975	98.4	erythroblast	58.5%
7	7184	75.4	myeloblast	99.1%
8	704	89.9	erythroblast	90.2%
9	5948	77.3	basophil	44.4%
10	4905	59.8	erythroblast	50.9%
11	3128	68.9	erythroblast	75.0%
12	2238	90.8	myeloblast	47.8%
13	10436	58.1	monocyte	81.1%
14	7513	64.1	myeloblast	86.8%
15	3630	65.1	monocyte	71.6%
16	4474	79.5	myeloblast	82.0%
17	2186	86.5	myeloblast	53.2%
18	2269	82.3	myeloblast	66.4%
19	5083	60.0	myeloblast	53.8%
20	8837	49.1	myeloblast	88.0%
21	2304	65.7	myeloblast	61.0%
22	4896	71.3	monocyte	61.7%
23	5621	53.6	erythroblast	53.8%
24	1095	106.2	myeloblast	99.8%
25	7869	69.3	myeloblast	98.0%
26	3563	64.2	myeloblast	50.4%
27	1163	79.6	myeloblast	45.7%
28	3603	65.1	monocyte	57.7%
29	1874	102.8	myeloblast	94.3%
30	3974	65.0	erythroblast	82.8%
31	2290	78.5	myeloblast	87.2%
32	3588	67.8	erythroblast	69.0%
33	4408	55.9	erythroblast	46.2%
34	2410	88.8	myeloblast	63.3%
35	4825	73.3	myeloblast	63.7%
36	4411	64.8	myeloblast	42.3%
37	2775	71.6	erythroblast	53.8%
38	3045	75.4	myeloblast	45.9%
39	4252	91.4	myeloblast	84.7%

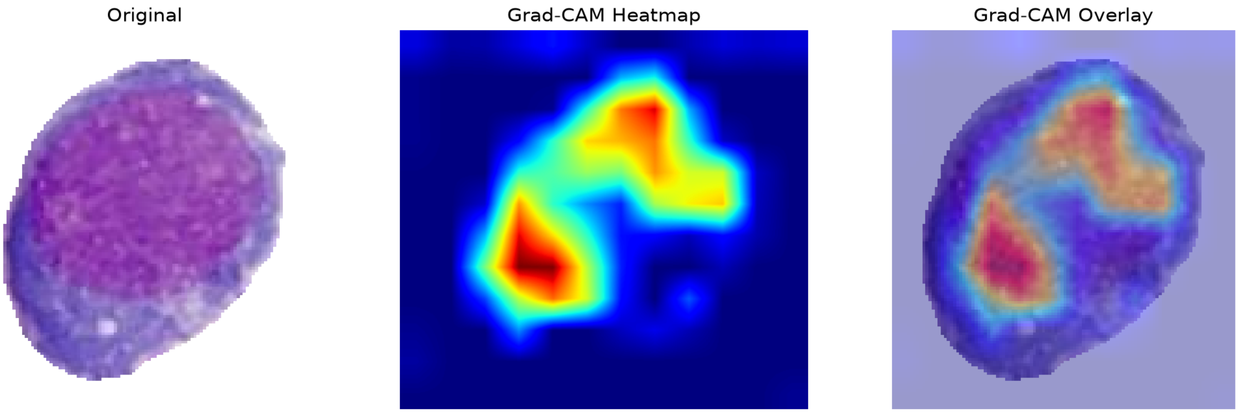
ID	Area	Purple	Class	Confidence
40	5281	82.4	myeloblast	90.5%
41	4235	57.9	erythroblast	62.9%
42	4079	61.6	monocyte	50.6%
43	5796	65.1	myeloblast	91.7%
44	7792	44.5	erythroblast	93.5%
45	3377	76.3	monocyte	50.1%
46	9041	71.4	myeloblast	99.4%
47	833	110.1	myeloblast	98.7%
48	5833	81.6	monocyte	59.9%
49	820	112.4	erythroblast	64.5%
50	7055	70.8	monocyte	74.6%
51	3618	65.1	erythroblast	56.4%
52	2760	102.9	myeloblast	90.2%
53	1700	111.5	myeloblast	91.1%
54	1475	80.8	erythroblast	68.0%
55	1941	110.5	myeloblast	83.3%
56	3372	69.8	myeloblast	89.3%
57	3523	79.0	myeloblast	96.1%
58	5194	78.0	monocyte	64.8%
59	4762	59.3	myeloblast	48.6%
60	1943	106.8	myeloblast	79.7%
61	6266	54.0	monocyte	46.6%
62	707	82.6	myeloblast	54.9%
63	2090	109.9	myeloblast	76.5%
64	1862	96.8	myeloblast	68.6%
65	1558	106.4	myeloblast	90.2%
66	6732	71.4	myeloblast	70.3%
67	6384	77.8	myeloblast	68.3%
68	5995	70.9	myeloblast	77.7%
69	2867	91.8	basophil	74.9%
70	4340	64.1	myeloblast	72.8%
71	7201	44.4	erythroblast	86.5%
72	3297	69.6	monocyte	48.4%
73	3306	69.0	erythroblast	79.2%
74	2506	100.9	myeloblast	78.7%
75	6122	81.7	myeloblast	68.0%
76	5942	75.7	myeloblast	92.0%
77	4347	60.9	erythroblast	88.0%
78	1411	122.6	myeloblast	75.8%
79	5769	60.4	monocyte	86.5%
80	1026	104.8	myeloblast	97.0%

ID	Area	Purple	Class	Confidence
81	4668	61.9	myeloblast	90.3%
82	3172	72.5	erythroblast	96.6%
83	2296	111.8	myeloblast	97.6%
84	4827	86.5	monocyte	77.8%
85	2060	116.8	myeloblast	99.1%
86	3132	73.0	myeloblast	90.9%
87	4292	62.3	monocyte	35.8%
88	3342	67.3	erythroblast	67.8%
89	6785	79.2	myeloblast	90.5%
90	4306	67.9	monocyte	89.8%
91	2766	95.0	myeloblast	75.7%
92	7084	70.0	myeloblast	82.0%
93	2149	93.0	monocyte	79.5%
94	2162	96.0	myeloblast	51.9%
95	2975	77.9	basophil	71.0%
96	1984	112.0	myeloblast	80.0%
97	1282	118.2	myeloblast	75.4%
98	4005	64.1	erythroblast	70.8%
99	9414	74.1	myeloblast	99.4%
100	7692	72.4	myeloblast	99.9%
101	6397	67.6	erythroblast	90.2%
102	5778	78.8	myeloblast	98.7%
103	4438	75.4	myeloblast	62.4%
104	10064	53.3	myeloblast	99.4%
105	4371	74.0	myeloblast	85.9%
106	4960	66.3	myeloblast	73.9%
107	6136	74.0	monocyte	75.7%
108	2747	80.6	myeloblast	49.1%
109	3302	75.9	erythroblast	81.9%
110	4523	62.7	erythroblast	63.0%
111	3775	105.8	myeloblast	97.0%
112	3172	120.5	myeloblast	96.4%
113	2134	62.3	erythroblast	55.3%
114	1603	63.7	erythroblast	92.2%
115	817	90.6	erythroblast	61.3%

Grad-CAM Explanations

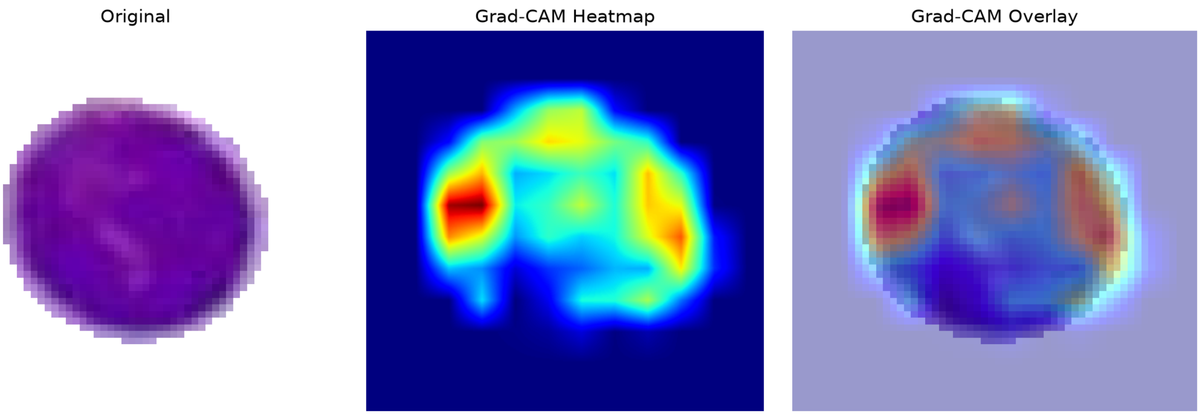
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 100 — myeloblast (99.9%)



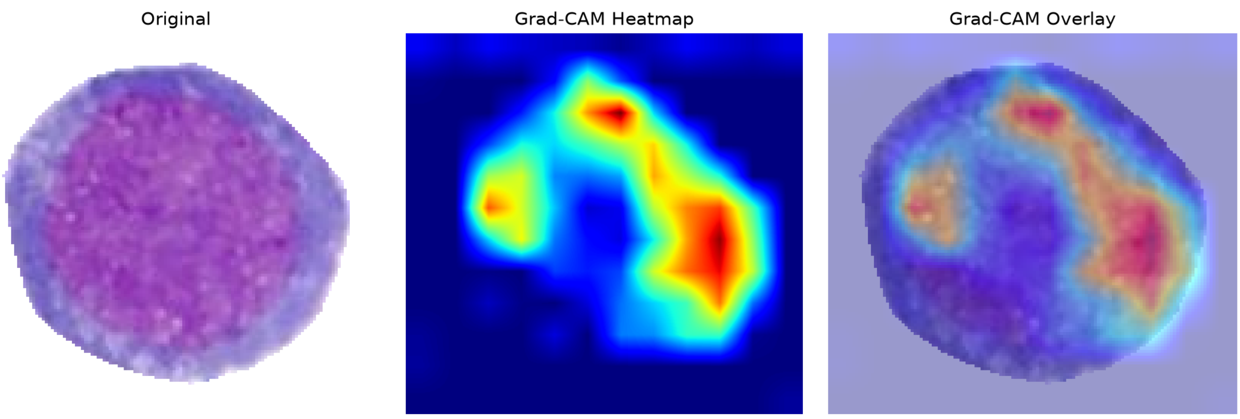
The ResNet50 model classified this segmented candidate as a myeloblast with 99.9% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 24 — myeloblast (99.8%)



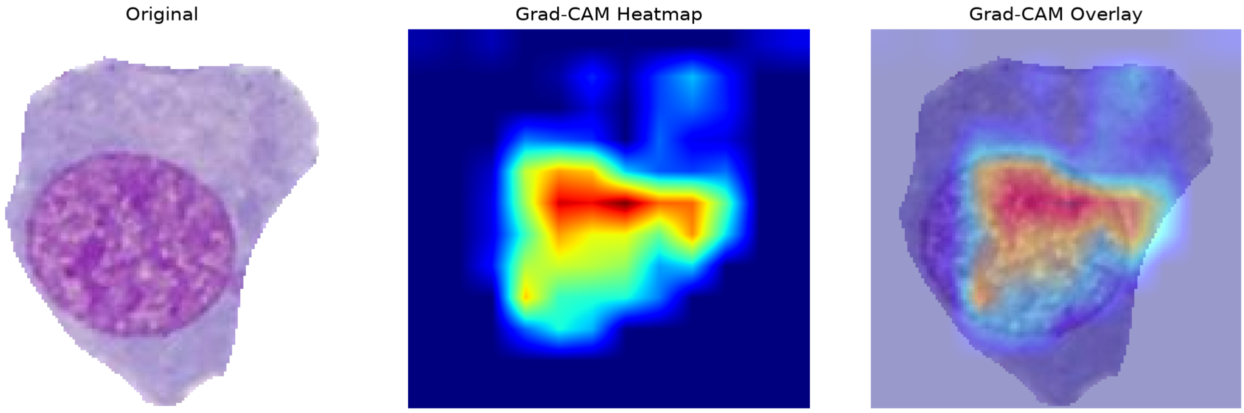
The ResNet50 model classified this segmented candidate as a myeloblast with 99.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 99 — myeloblast (99.4%)



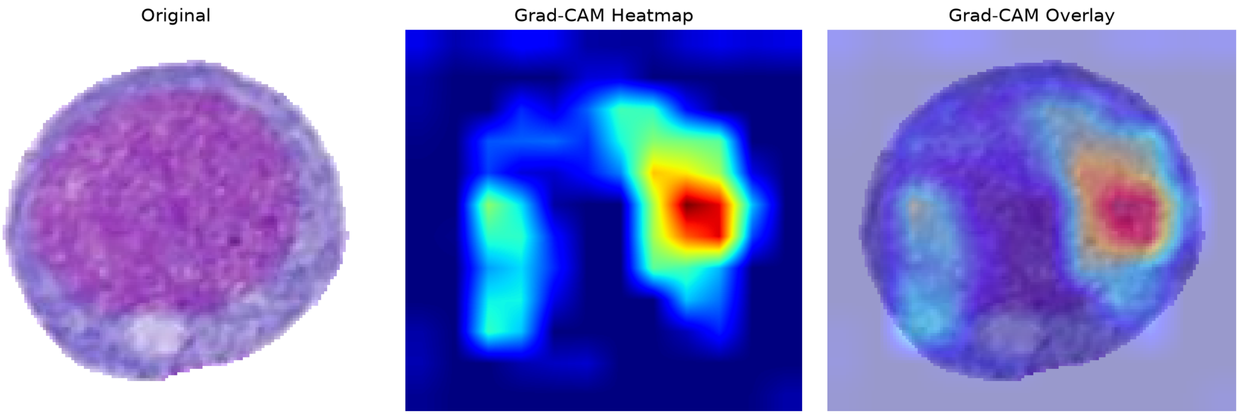
The ResNet50 model classified this segmented candidate as a myeloblast with 99.4% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 104 — myeloblast (99.4%)



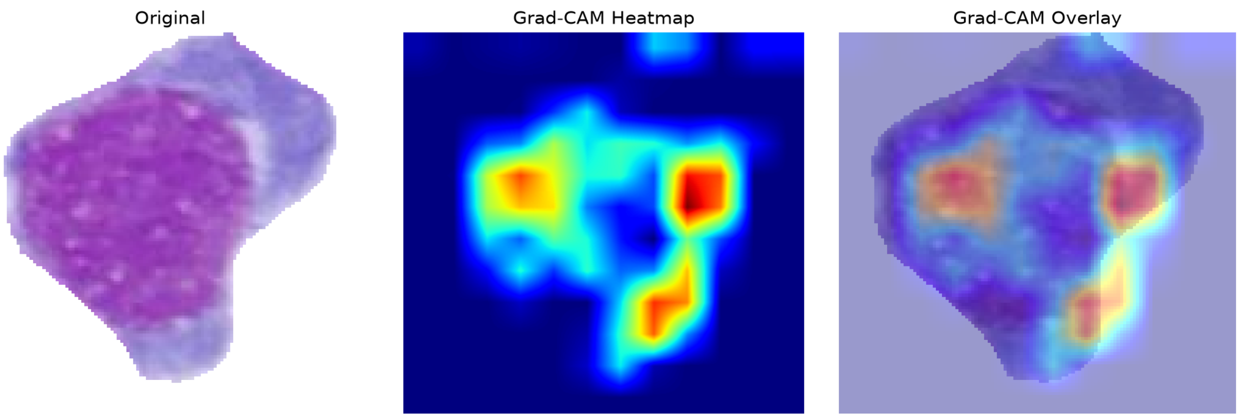
The ResNet50 model classified this segmented candidate as a myeloblast with 99.4% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 46 — myeloblast (99.4%)



The ResNet50 model classified this segmented candidate as a myeloblast with 99.4% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 7 — myeloblast (99.1%)



The ResNet50 model classified this segmented candidate as a myeloblast with 99.1% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

- The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.
- RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.
- Cellpose segmentation quality affects every downstream result.
- Stain, microscope, illumination and acquisition differences can cause domain shift.
- Confidence is not a calibrated probability of disease.
- A myeloblast-like prediction is not equivalent to an AML diagnosis.
- Clinical assessment requires appropriate hematopathology and laboratory testing.